WORKED SOLUTION

STEP II 2006, Question 8

Pure • Vector Geometry of Lines

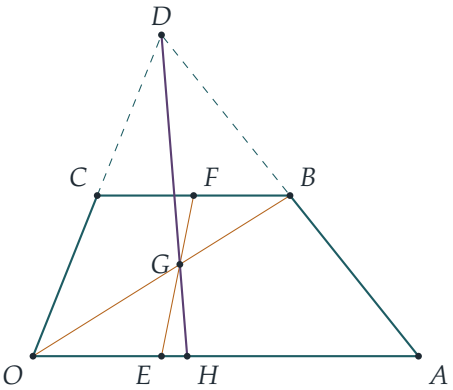
STEP II 2006, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q8

The key idea

Every point in the question is defined as the intersection of two lines, so the whole question runs on one routine: write each line in the form from the first part, set the two expressions equal, and compare coefficients of $\mathbf{a}$ and $\mathbf{c}$. That comparison is legitimate because $\mathbf{a}$ and $\mathbf{c}$ are not parallel, so a vector can be written as $p\mathbf{a} + q\mathbf{c}$ in only one way.



Solution

The equation of a line

The line through the points with position vectors $\mathbf{x}$ and $\mathbf{y}$ passes through $\mathbf{x}$ and has direction $\mathbf{y} - \mathbf{x}$, so its points are

$$\mathbf{r} = \mathbf{x} + \alpha(\mathbf{y} - \mathbf{x}) = (1 - \alpha)\mathbf{x} + \alpha\mathbf{y},$$

as α runs through all real numbers. Note that $\alpha = 0$ gives $\mathbf{x}$, $\alpha = 1$ gives $\mathbf{y}$, and the coefficients always add up to 1. That is the required form.

Throughout, $\mathbf{a}$ and $\mathbf{c}$ are not parallel (otherwise O, A, C would be collinear and there

would be no trapezium), so if $p\mathbf{a}+q\mathbf{c}=p'\mathbf{a}+q'\mathbf{c}$ then $p=p'$ and $q=q'$.

Part (i)

Since CB is parallel to OA, the vector $\overrightarrow{CB}$ is a scalar multiple of $\overrightarrow{OA}=\mathbf{a}$; call it $\lambda\mathbf{a}$. Then

$$\mathbf{b}=\overrightarrow{OC}+\overrightarrow{CB}=\mathbf{c}+\lambda\mathbf{a}.$$

Going round $OABC$ in order, $\overrightarrow{CB}$ points the same way as $\overrightarrow{OA}$, so $\lambda>0$; and $CB<OA$ gives $\lambda<1$. So $0<\lambda<1$, and in particular $\lambda\neq 1$, which we shall need.

Part (ii)

E **and** F**.** These come straight from the ratios:

$$\mathbf{e}=\tfrac{1}{3}\mathbf{a},\qquad \mathbf{f}=\tfrac{1}{2}(\mathbf{c}+\mathbf{b})=\tfrac{1}{2}\lambda\mathbf{a}+\mathbf{c}.$$

D**.** Points of line OC are $\mu\mathbf{c}$. Points of line AB are

$$(1-\beta)\mathbf{a}+\beta(\lambda\mathbf{a}+\mathbf{c})=(1-\beta+\beta\lambda)\mathbf{a}+\beta\mathbf{c}.$$

At D these agree. Comparing coefficients gives $\beta=\mu$ and $1-\beta(1-\lambda)=0$, so $\beta=\dfrac{1}{1-\lambda}$ (fine, since $\lambda\neq 1$). Hence

$$\mathbf{d}=\frac{1}{1-\lambda}\,\mathbf{c}.$$

Since $\frac{1}{1-\lambda}>1$, D does lie on OC produced beyond C, as it should.

G**.** Points of OB are $t(\lambda\mathbf{a}+\mathbf{c})$. Points of EF are

$$(1-s)\tfrac{1}{3}\mathbf{a}+s(\tfrac{1}{2}\lambda\mathbf{a}+\mathbf{c})=(\tfrac{1}{3}-\tfrac{1}{3}s+\tfrac{1}{2}\lambda s)\mathbf{a}+s\mathbf{c}.$$

Comparing $\mathbf{c}$-coefficients gives $s=t$; then the $\mathbf{a}$-coefficients give

$$\lambda t=\tfrac{1}{3}-\tfrac{1}{3}t+\tfrac{1}{2}\lambda t\quad\Longrightarrow\quad t\Big(\frac{3\lambda+2}{6}\Big)=\frac{1}{3}\quad\Longrightarrow\quad t=\frac{2}{3\lambda+2}.$$

So

$$\mathbf{g}=\frac{2}{3\lambda+2}\,(\lambda\mathbf{a}+\mathbf{c}).$$

H**.** Points of line DG are

$$(1-k)\mathbf{d}+k\mathbf{g}=\frac{2k\lambda}{3\lambda+2}\,\mathbf{a}+\Big(\frac{1-k}{1-\lambda}+\frac{2k}{3\lambda+2}\Big)\mathbf{c}.$$

Ask yourself How do we pick out the point of this line that lies on OA?

Points of line OA are multiples of $\mathbf{a}$ alone, so we need the $\mathbf{c}$-coefficient to vanish. Multiplying by $(1-\lambda)(3\lambda+2)$,

$$\begin{aligned}(1-k)(3\lambda+2)+2k(1-\lambda)&=0\\(3\lambda+2)-k(3\lambda+2-2+2\lambda)&=0\\k&=\frac{3\lambda+2}{5\lambda}.\end{aligned}$$

The $\mathbf{a}$-coefficient is then

$$\frac{2\lambda}{3\lambda+2}\cdot\frac{3\lambda+2}{5\lambda}=\frac{2}{5},\qquad\text{so}\qquad\mathbf{h}=\tfrac{2}{5}\,\mathbf{a}.$$

Collecting the answers:

$$\mathbf{d}=\frac{\mathbf{c}}{1-\lambda},\quad\mathbf{e}=\tfrac{1}{3}\mathbf{a},\quad\mathbf{f}=\tfrac{1}{2}\lambda\mathbf{a}+\mathbf{c},\quad\mathbf{g}=\frac{2(\lambda\mathbf{a}+\mathbf{c})}{3\lambda+2},\quad\mathbf{h}=\tfrac{2}{5}\mathbf{a}.$$

Since $OH=\frac{2}{5}OA$ and $HA=\frac{3}{5}OA$,

$$OH:HA=2:3,$$

whatever the shape of the trapezium.

Checks

- Take $\mathbf{a}=(6,0)$, $\mathbf{c}=(1,2.5)$ and $\lambda=\frac{1}{2}$ (the diagram). The formulas give $D=(2,5)$, $G\approx(2.286,1.429)$ and $H=(2.4,0)$; the slope from D to G is $-3.571/0.286=-12.5$, and so is the slope from D to H, namely $-5/0.4$. So D, G, H really are collinear.
- The answer for H doesn't depend on λ, $\mathbf{a}$ or $\mathbf{c}$. A computer repeated the whole construction in exact rational arithmetic for 2000 random trapezia and found $OH:HA=2:3$ every time.
- As $\lambda\to1$ the sides OC and AB become parallel, and $\mathbf{d}=\mathbf{c}/(1-\lambda)$ runs off to infinity, as it should.

Common slips

- Using the same parameter letter for two different lines. Each line needs its own parameter before you set them equal.
- Comparing coefficients without saying why it's allowed: it works because $\mathbf{a}$ and $\mathbf{c}$ are not parallel.
- Writing $\mathbf{b}=\mathbf{a}+\lambda\mathbf{c}$. It is $\overrightarrow{CB}$, not $\overrightarrow{AB}$, that is parallel to $\mathbf{a}$.

Beyond STEP

The answer $2:3$ doesn't depend on the shape of the trapezium because every step uses only lines, intersections and ratios along lines, and an affine map (a linear map

followed by a translation) preserves all of these. So you could send the trapezium to a convenient one, say with $OABC$ having C directly above O, and compute there. This is the idea of *affine geometry*; the coefficient form $(1-\alpha)\mathbf{x}+\alpha\mathbf{y}$ is the start of barycentric coordinates. Strang's *Introduction to Linear Algebra* covers linear maps and coordinates, and Siklos's *Advanced Problems in Mathematics* has more vector geometry questions in this style.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.